

WEEK 8

Project Report: Student Performance Dashboard

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1. Introduction

The goal of this project was to create a two-page interactive dashboard in Power BI to analyze student grades and see how factors like test prep, parental education, and lunch types affect performance.

Main Dashboard

- KPI Cards: Created cards to show key metrics, including Average Math Score, Average Reading Score, Average Writing Score, Overall Average Score, and Overall Pass % (formatted as a percentage).
- Analytical Charts:
 - Created a horizontal bar chart to see scores by Race/Ethnicity.
 - Created a vertical column chart to compare scores by Test Preparation Course.
 - Created a pie chart to analyze scores by Parental Level of Education.
 - Created a vertical column chart to evaluate scores by Lunch Type.
 - Created a donut chart to show the overall Grade Distribution (A, B, C, F).
- Interactive Slicers: Added three filters at the bottom so users can filter the entire page by Race/Ethnicity, Lunch Type, and Parental Education.

Academic Profile (Drill-Through Page)

- Created a second page and set up a Drill-Through feature using the Grade field. This automatically added a Back Button (←) to navigate back to Page
- Added a Math Average Card.
- Created a Pie Chart comparing all three subject scores together.
- Created a Funnel Chart showing overall scores by Gender.
- Created a vertical Column Chart at the bottom showing performance by both Gender and Lunch Type.

Key Insights Found

1. Students who completed test preparation scored higher in all subjects.
2. Students with Standard lunch options had higher average scores than those with Free/Reduced lunch.
3. Higher parental education levels strongly connect to higher student test scores.
4. Overall, student performance was highest in Reading and lowest in Math.
5. Female students achieved a higher overall score average than male students.

Dashboard

